

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and implemented **data pipelines** processing **tens of billions of events** daily using **Python**, **SQL**, and **AWS** (EC2, S3, Lambda) for enterprise payment-processing systems; deployed via **Jenkins CI/CD** with **GitHub** version control, achieving **99.9% SLA uptime** in production environments.
- Built **scalable data engineering infrastructure** supporting annotation systems and data quality frameworks; developed **ETL pipelines** with **Python** and **SQL** (PostgreSQL, MySQL, Oracle) to ingest, validate, and transform structured financial and operational data across interconnected cloud systems at millisecond latencies.
- Developed **data visualization dashboards** and operational reporting tools using **Power BI** and **Tableau** to surface model quality metrics, data biases, and error patterns; collaborated with data scientists, engineers, and domain taxonomists to translate complex datasets into actionable insights for stakeholder decision-making.
- Engineered **API/SDK integrations** between **cloud-based systems** and third-party platforms; built **REST APIs** in **Java Spring Boot** and developed **Python**-based integration tools handling **hundreds of thousands of transactions per second** with robust error handling and audit logging for regulatory compliance.
- Applied **software engineering rigor** across data pipeline development through **CI/CD automation** (Jenkins), code reviews, **Git** branching strategies, and containerization with **Docker**; implemented structured logging, monitoring via **CloudWatch** and **Splunk**, and incident response protocols ensuring data integrity and system reliability.
- Conducted statistical analysis and data quality investigations using **Python** (pandas, NumPy, scikit-learn) and **R**; identified annotation biases, error patterns, and data drift in large-scale datasets; produced technical documentation and trained cross-functional teams on data inspection methodologies and tool usage in Agile/Scrum environments.

SKILLS

Data Engineering & Pipelines – Python, SQL (PostgreSQL, MySQL, Oracle), ETL design, Apache Spark, data pipeline architecture, cloud-based data integration, REST APIs, SDK interaction.

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Databricks, Docker, Linux, shell scripting, CI/CD (Jenkins, GitHub Actions), containerization.

Data Quality & Visualization – Power BI (dashboards, DAX, data modelling), Tableau, data inspection tools, bias detection, error pattern analysis, operational reporting.

Machine Learning & Analytics – scikit-learn, supervised/unsupervised learning, statistical analysis, hypothesis testing, model evaluation metrics, A/B testing, Big Data frameworks (Hadoop, Spark).

Software Engineering Practices – CI/CD automation, Git version control, code review disciplines, Agile/Scrum methodologies, software architecture, database design, scalable systems development.

Domain & Collaboration – Cross-functional team coordination (data scientists, engineers, taxonomists, knowledge engineers), annotation systems support, change-management, regulatory compliance (GDPR, audit trails), technical documentation.

Languages & Certifications – Python, Java, R, SQL; IELTS Academic Band 7; Master's degree (Data Analytics); Full Stack development (Java Spring Boot, Angular, React).

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin

02/2026

BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India

01/2021

PROJECTS

Real-Time Payment Operations Data Pipeline (HCL Technologies)

- Designed and deployed a **data pipeline** ingesting SWIFT messaging events and payment transactions into a centralized data warehouse; processed **100+ million records daily** using **Python ETL scripts**, **SQL** transformations, and **AWS Lambda** for event-driven processing, reducing data latency from **2 hours to 15 minutes**.
- Built **Power BI dashboards** for transaction-flow visibility and operational alerting; implemented **DAX measures** and **Power Query** ETL to surface payment-processing KPIs, SLA compliance metrics, and error patterns; presented model quality insights to banking operations teams across multiple time zones.
- Integrated third-party API endpoints using **REST API calls** and **Python SDK** interactions; managed **cloud-based system** data flows via **AWS S3** and **EC2**; implemented robust **CI/CD pipelines** with **Jenkins** and **GitHub** to automate data validation, schema migrations, and deployment, achieving zero-downtime releases.

Ireland Jobs Tracker – AI-Driven Resume Generation & Workflow Automation (Personal Project)

- Built end-to-end **data pipeline** orchestrating job description scraping, ingestion, and processing; integrated **Claude API** (Anthropic) for intelligent resume generation using **Python** and structured JSON prompt engineering with constraint guardrails; deployed scalable workflow across **Apify scrapers**, **AWS** compute, and **GitHub Actions CI/CD**.
- Engineered **data quality framework** to validate scraped job postings, detect biases in JD terminology, and normalize structured data; built **error handling and logging** infrastructure using **Python** to surface data anomalies; created analytical reports identifying trends across job market datasets.
- Developed **data visualization outputs** (reportlab PDF generation) and published dashboards on **Vercel**; implemented **CI/CD automation** with **GitHub Actions** for continuous data pipeline execution; applied **software engineering practices** (modular design, testing, documentation) to maintain reliability and scalability.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)